0.1 Lipschitz Condition

In this section, X and Y are assumed to be Metric Spaces.

Definition 0.1.1. Let (X, d_X) and (Y, d_Y) be Metric Spaces.
A function $f : X \to Y$ is said to be *Lipschitz Continuous* if: there exists $L > 0$ such that for any $x_1, x_2 \in X$,

$$d_Y(f(x_1), f(x_2)) \leq L \cdot d_X(x_1, x_2)$$

The constant L is called *Lipschitz Constant* of f.

Lemma 0.1.2. Let $f : X \to Y$ be a Lipschitz function. Define a *Dilation* of f as:

$$D \stackrel{\text{def}}{=} \sup_{x_1 \neq x_2} \frac{d_Y(f(x_1), f(x_2))}{d_X(x_1, x_2)}$$

Then,

$$D = \min\{L > 0 \mid L \text{ is a Lipschitz Constant of } f\}$$

Proof. Let $x_1, x_2 \in X$ be given with $x_1 \neq x_2$ (If $x_1 = x_2$, then there is nothing to prove.) Then, by definition,

$$\frac{d_Y(f(x_1), f(x_2))}{d_X(x_1, x_2)} \leq D$$

That is,

$$d_Y(f(x_1), f(x_2)) \leq D \cdot d_X(x_1, x_2)$$

Thus, the Dilation is also Lipschitz constant.
Meanwhile, if $L > 0$ is a Lipschitz Constant of f, this means f is an upper bound of a set:

$$\left\{ \frac{d_Y(f(x_1), f(x_2))}{d_X(x_1, x_2)} \,\middle|\, x_1, x_2 \in X, x_1 \neq x_2 \right\}$$

By definition of Supremum, $D \leq L$. As a result, we obtain the conclusion. $\square$

0.1.1 Properties

Proposition 0.1.3. If $f : X \to Y$ is Lipschitz Continuous, then f is uniformly continuous.

Proof. Let $L \geq 0$ be a Lipschitz Constant of f. Then, for any $\varepsilon > 0$,

$$\forall x_1, x_2 \in X,\ d_X(x_1, x_2) < \frac{\varepsilon}{L} \implies d_Y(f(x_1), f(x_2)) \leq L \cdot d_X(x_1, x_2) < \varepsilon$$

□

Note that:

$$\text{Lipschitz Continuous} \implies \text{Uniformly Continuous} \implies \text{Continuous}$$

Proposition 0.1.4. Let $f : (a, b) \to \mathbb{R}$ be a Differentiable function where (a, b) is an interval. Then,

f is Lipschitz Continuous *if and only if* f' is bounded in (a, b).

Proof.
$\implies$)
Let $L > 0$ be a Lipschitz constant of f, and $x \in (a, b)$ be given. Since definition of derivative,

$$f'(x) \overset{\text{def}}{=} \lim_{t \to x} \frac{f(x) - f(t)}{x - t}$$

Meanwhile, the assumption gives: for any distinct $x, t \in (a, b)$,

$$\frac{|f(x) - f(t)|}{|x - t|} \leq L$$

Therefore,

$$f'(x) = \lim_{t \to x} \frac{f(x) - f(t)}{x - t} \leq \lim_{t \to x} \frac{|f(x) - f(t)|}{|x - t|} \leq \lim_{t \to x} L = L$$

$\impliedby$)
Let distinct $x, y \in (a, b)$ be given. Then, the Mean-Value Theorem gives: There exists a $z \in (x, y)$ such that

$$f(x) - f(y) = f'(z)(x - y) \implies f'(z) = \frac{f(x) - f(y)}{x - y}$$

Now,

$$\left| \frac{f(x) - f(y)}{x - y} \right| = |f'(z)| \leq L \implies |f(x) - f(y)| \leq L \cdot |x - y|$$

If $x = y$, then there is nothing to prove. □

0.1.2 Banach Fixed Point Theorem

Definition 0.1.5. Let $f: X \to X$ be any function. A point $x \in X$ is called a *fixed point* of f if $f(x) = x$.

Definition 0.1.6. A Lipschtiz function $f: X \to X$ is called *Contractive* with respect to the metric d if:

There exists a Lipschitz Constant L of f such that $0 < L < 1$

Theorem 0.1.7. *Banach Fixed point Theorem*
Suppose (X, d) is a Complete Metric Space, and $f: X \to X$ is a Contractive map.
Then, there exists a unique fixed point of f, $x^* \in X$.

Proof. Let $\alpha \in (0,1)$ be a Lipschitz Constant of f. Clearly,

$$\text{Contractive} \implies \text{Lipschitz} \implies \text{Continuous.}$$

Thus, f is Continuous.
Let $x_0 \in X$ be arbitrary, and construct a sequence $\{x_n\}$ recursively as follows:

$$x_{n+1} \overset{\text{def}}{=} f(x_n), \;\; n \geq 0$$

Then, for any $n \geq 0$,

$$\begin{aligned} d(x_{n+1}, x_n) &\overset{\text{def}}{=} d(f(x_n), f(x_{n-1})) \leq \alpha d(x_n, x_{n-1}) \\ &\overset{\text{def}}{=} \alpha d(f(x_{n-1}), f(x_{n-2})) \leq \alpha^2 d(x_{n-1}, x_{n-2}) \\ &\;\;\vdots \\ &\overset{\text{def}}{=} \alpha^{n-1} d(f(x_1), f(x_0)) \leq \alpha^n d(x_1, x_0) \end{aligned}$$

Let $\varepsilon > 0$ be given. Put $N \in \mathbb{N}$ such that $\alpha^N \cdot d(x_1, x_0) < \varepsilon(1-\alpha)$. Then, $n \geq m \geq N$ implies that

$$\begin{aligned} d(x_n, x_m) &\leq d(x_n, x_{n-1}) + d(x_{n-1}, x_{n-2}) + \cdots + d(x_{m+1}, x_m) \\ &\leq \alpha^n d(x_1, x_0) + \alpha^{n-1} d(x_1, x_0) + \cdots + \alpha^{m+1} d(x_1, x_0) \\ &= \alpha^{m+1} d(x_1, x_0) \sum_{r=0}^{n-m-1} \alpha^r < \alpha^N d(x_1, x_0) \sum_{r=0}^{\infty} \alpha^r < \varepsilon \cdot (1-\alpha) \cdot \frac{1}{1-\alpha} = \varepsilon \end{aligned}$$

Therefore, $\{x_n\}$ is Cauchy sequence. Since X is Complete, for some $x^* \in X$, $\lim\limits_{n\to\infty} x_n = x^*$. Consequently,

$$f(x^*) = f\left(\lim_{n\to\infty} x_n\right) \overset{f \text{ conti.}}{=} \lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} x_{n+1} = x^*$$

Uniqueness is given by: if $x_1, x_2 \in X$ are fixed point of f, then

$$d(x_1, x_2) = d(f(x_1), f(x_2)) \leq \alpha \cdot d(x_1, x_2)$$

But, since $\alpha < 1$, $d(x_1, x_2)$ must be zero. That is, $x_1 = x_2$. □